

ASSIGNMENT - I

I. Inverse Laplace Transforms of Standard Functions, Linearity Property :

Find the Inverse Laplace Transforms of the following :

1. $L^{-1} \left[\frac{3s-12}{s^2+8} \right]$

Ans : $3\cos 2\sqrt{2}t - 3\sqrt{2}\sin 2\sqrt{2}t$

2. $L^{-1} \left[\frac{5s+10}{9s^2-16} \right]$

Ans : $\frac{5}{9} \cosh \left(\frac{4}{3} \right) t + \frac{5}{6} \sinh \left(\frac{4}{3} \right) t$

3. $L^{-1} \left[\frac{3(s^2-2)^2}{2s^5} \right]$

Ans : $\frac{3}{2} - 3t^2 + \frac{t^4}{4}$

4. $L^{-1} \left[\frac{6}{2s-3} - \frac{3+4s}{9s^2-16} + \frac{8-6s}{16s^2+9} \right]$

Ans : $3e^{(3/2)t} - \frac{1}{4} \sinh \left(\frac{4}{3} \right) t - \frac{4}{9} \cosh \left(\frac{4}{3} \right) t + \frac{2}{3} \sin \left(\frac{3}{4} \right) t - \frac{3}{8} \cos \left(\frac{3}{4} \right) t$

5. $L^{-1} \left[\frac{1}{s} \sin \left(\frac{1}{s} \right) \right]$ {Use the series expansion of $\sin x$ }

Ans : $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} t^{2n-1}}{[(2n-1)!]^2}$